Course Title	Basic Mechanical Engineering	**Course Code**	ME2001
Department	Department of Electrical Engineering (DEE)	**Campus**	Lahore
Knowledge Profile	Engineering Fundamentals (WK3)	**Credit Hrs.**	3
Knowledge Area	Interdisciplinary Engineering (KA09)	**Grading Scheme**	Relative
HEC Knowledge Area	Inter Disciplinary Engg. Breadth (Elective)	**Applicable From**	Spring 2023
Pre-requisite(s)			

Course Objective	To build the fundamental concepts of engineering mechanics and to clearly present their application in the analysis of structures and planner motion of mechanical components.

No.	Assigned Program Learning Outcome (PLO)
02	An ability to identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.

I = Introduction, R = Reinforcement, E = Evaluation.
A = Assignment, Q = Quiz, M = Midterm, F = Final, L = Lab, P = Project, W = Written Report.

No.	Course Learning Outcome (CLO) Statements	Assessment Tools	Taxonomy Levels	PLO
01	Calculate the moment of a force/couple	A1, M1	C2	2
02	Analyze static equilibrium analysis of a rigid body by applying Newton Laws of Motion and concept of dry friction.	Q1, M1	C4	2
03	Analyze absolute motion of rigid bodies in general plane motion	Q2, M2	C4	2
04	Analyze relative motion of rigid bodies in rotation and/or translation	Q3, M2	C4	2
05	Evaluate the internal forces in the members of a loaded truss and assess safety of the structure.	A2, F	C6	2

Text Book(s)	**Title**	Engineering Mechanics: Statics and Dynamic, 12th Ed
	Author	R. C. Hibbeler
	Publisher	Pearson Prentice Hall
Ref. Book(s)	**Title**	Statics and Mechanics of Materials,4th Ed
	Author	R. C. Hibbeler
	Publisher	Pearson Prentice Hall
	Title	Engineering Mechanics: Statics and Dynamics
	Author	Meriam, J. L. and Kraige, L. G
	Publisher	John & Wiley Sons

Week	Course Contents/Topics	Chapter*	CLO*
1	General Principles	1	1
2	Introduction to Vector Mechanics	2	1
3	Force Systems	4.1 to 4.6	1
4	Free Body Diagram	3.2, 5.2	1
5	Equilibrium of Rigid Body	5.1, 5.3 to 5.6	2
6	Dry Friction	8.1, 8.2	2
7	Planar Kinematics: Absolute Motion Analysis	16.1 to 16.3	3
8	Planar Kinematics: Absolute Motion Analysis	16.4	3
9	Planar Kinematics: Relative Motion Analysis	16.5	4
11	Planar Kinematics: Relative Motion Analysis	16.7	4
12	Mechanics of Materials: Stress and Strain	7*	5
13	Mechanics of Materials: Torsion and Bending	10*,11*	5
14	Mechanical Properties of Material	8*	5
15	Structure Analysis: Method of Joints	6	5
16	Structure Analysis: Method of Sections	6	5

***Reference book chapters are given in brackets**

Assessment Tools	Weightage
Quizzes (3), Assignments (3)	20.0%
Midterm (1+2)	30.0%
Final Exam	50.0%